

Database

Credentialed Access

INSPIRE, a publicly available research dataset for perioperative medicine

Leerang Lim , Hyung-Chul Lee

Published: Aug. 12, 2024. Version: 1.3

When using this resource, please cite:

“ Cite

Copy BibTeX

Lim, L., & Lee, H. (2024). INSPIRE, a publicly available research dataset for perioperative medicine (version 1.3). *PhysioNet*. RRID:SCR_007345. <https://doi.org/10.13026/46m4-f655>

Additionally, please cite the original publication:
[Leerang Lim et al. INSPIRE, a publicly available research dataset for perioperative medicine. Sci Data. 2024 Jun 21;11\(1\):655.](#)

Please include the standard citation for PhysioNet: [\(show more options\)](#)
Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., ... & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation* [Online]. 101 (23), pp. e215–e220. RRID:SCR_007345.

Abstract

We present the INSPIRE dataset, a publicly available research dataset in perioperative medicine, which includes approximately 130,000 cases (50% of all surgical cases) who underwent anesthesia for surgery at an academic institution in South Korea between 2011 and 2020. This comprehensive dataset includes patient characteristics such as age, sex, American Society of Anesthesiologists physical status classification, diagnosis, surgical procedure code, department, and type of anesthesia. It also includes vital signs in the operating theatre, general wards, and intensive care units (ICUs), laboratory results from six months before admission to six months after discharge, and medication during hospitalization. Complications include total hospital and ICU length of stay and in-hospital death. We hope this dataset will inspire collaborative research and development in perioperative medicine and serve as a reproducible external validation dataset to improve surgical outcomes.

Background

Major complications after surgery occur in approximately 7-15% of patients [1]. However, thorough research on rare complications, such as postoperative mortality, respiratory failure, or myocardial injury, requires a comprehensive large dataset for adequate statistical power. There are large registries such as the National Anesthesia Clinical Outcomes Registry or the National Surgical Quality Improvement Program are national-wide programs to improve outcomes in surgical patients [2-3]. However, these data are only available to researchers at participant institutions and do not include detailed data such as time series of laboratory or physiological measurements.

The VitalDB, the largest publicly available intraoperative dataset for surgical patients, provides high-resolution multi-parameter data [4-5]. However, it still only includes 57 cases (0.9%) of in-hospital mortality from 6,388 cases at a single center in South Korea. The Medical Information Mart for Intensive Care (MIMIC), another publicly available dataset of ICU patients from a single center in the United States, provides a more extensive range of patients with complications. However, MIMIC is limited to a specific cohort of patients admitted to either the intensive care unit (ICU) or emergency department [6].

In recent years, a large number of machine learning models have been introduced with the aim of improving risk stratification, predicting adverse events, and alerting to deterioration in perioperative medicine [7-9]. However, a prevailing issue with these models is the lack of external validation, which hinders their unbiased, objective performance evaluation prior to their implementation in clinical practice [10]. The creation of an open dataset can play a pivotal role in this situation by providing the research community with an objective validation set, accelerating technology development through collaboration, and advancing medical knowledge by reducing the disparities in the accessibility of clinical data.

Here, we present a collaborative research dataset called INSPIRE, an INformative Surgical Patient dataset for Innovative Research Environment, which contains sufficient data for collaborative research and development in perioperative medicine. The primary purpose of this dataset is to facilitate the development of novel predictive models and to serve as an external validation resource for existing models. By enabling such research efforts, we hope to ‘inspire’ innovative research in perioperative medicine and improve surgical patient outcomes.

Methods

Ethical approval

This study was approved by the Institutional Review Board (IRB) of Seoul National University Hospital (SNUH, IRB No. H-2210-078-1368). The IRB also waived the informed consent due to the retrospective nature of the study design. Additionally, the Institutional Data Review Board (DRB) of SNUH approved the release of the dataset to the public after a review of the dataset with the decision of adequate de-identification (BRB No. BD-R-2022-11-02).

Patient population

All patients who received surgery under general, neuraxial, regional, and monitored anesthesia care between January 2011 and December 2020 at SNUH were included. Patients younger than 18 or older than 90 on the operation day and patients, who refused to disclose their admission, or disclosed to the public, such as articles on mass media, were excluded. After the exclusions, 50% of the patients were randomly selected to populate the publicly released dataset.

Data acquisition

The operation and anesthesia-related variables, diagnosis, vital signs, laboratory results, or prescription and administration of the medications were extracted from the clinical data warehouse of the SNUH (SUPREME version 1.0 and 2.0).

The patient’s vital signs and the anesthesia machine settings in the operating room were recorded automatically on the anesthesia records every 1 minute. The anesthesia records include manual recordings of urine output, estimated blood loss, infused fluid or blood product volume, and values of specialized monitoring devices, such as processed electroencephalogram or pulmonary artery catheter, as a free-text form.

Throughout the duration of a patient's occupancy in the Intensive Care Unit (ICU), a range of parameters are documented either hourly or at intervals stipulated by the attending clinicians. These parameters encompass vital signs, urine output, volumes of infused fluids or blood products, as well as metrics obtained from the mechanical ventilator. The vital signs and mechanical ventilator-derived variables were acquired through digital communication with the respective equipment allowing manual modification. Values regarding additional life-supporting devices such as continuous renal replacement treatment (CRRT), extracorporeal membrane oxygenation (ECMO), or intra-aortic balloon pump (IABP) were recorded per 4~8 hours as a free-text form and converted to the binary variable.

During the patient’s stay in the general ward, vital signs were measured and recorded 4~6 times per day according to the physician’s order. Diagnoses were recorded according to the International Classification of Diseases, 10th revision, Clinical Modification (ICD-10-CM) [11]. Operation names were extracted from the operation records and the data of claims for the National Health Insurance Service as forms of free text [12].

Data processing

All vital signs, laboratory results, and the use of specialized devices, such as mechanical ventilators, CRRT, ECMO, and IABP, were aggregated to the median value with the maximal resolution of 5 minutes. For the use of the specialized devices, we extracted the start time of using the devices. If there was no record of more than 4 or 8 hours (99% of the distribution of record interval for mechanical ventilation and CRRT), it was considered the end time of using the devices. Laboratory test results and vital signs were extracted for values measured between 30 days before and after each operation. Types of laboratory data included arterial blood gas analysis, blood cell count, renal and liver function test, coagulation tests, hemoglobin A1c, lactate, and cardiac enzymes.

Anonymization

Data were extracted on a subject basis. For each subject, the initial admission time for operation during the study period was regarded as time zero. All times were converted to times (minutes) relative to the time zero.

According to the Personal Information Protection Act of South Korea (2020 revision), all personal information that can identify an individual by itself or when combined with other information, such as names, identification numbers (resident registration number, passport number, insurance number, etc.), address, and telephone number should be removed before sharing. We did not extract these variables except for the medical record number, which was subsequently substituted with unique random

integers as the subject id (described as *subject_id*). Each admission and operation were randomly assigned with unique numbers (*hadm_id* and *op_id*, respectively). To allow identifiers to be easily distinguished, *subject_id* begin with “1”, *hadm_id* begin with “2”, and *op_id*, respectively begin with “4”.

A list of patients who opted out of data sharing was obtained from the office for hospital information at SNUH, and 16,176 patients with 25,946 operations cases were identified for exclusion. To exclude patients who were identifiable by widely accessible data such as public media, we searched using Google News with the keywords (“Seoul National University Hospital” or “SNUH” and “operation” or “procedure”) and removed 3 patients with 3 cases.

Variables measuring the clinical state through physical and chemical methods, such as weight, height, vital signs, and laboratory results, were categorized into the 5th percentile values for anonymity. For each measured variable, the values below the 2.5th percentile were replaced by the 2.5th percentile value, and the values above the 97.5th percentile were replaced by the 97.5th percentile value. The rest were replaced with the closest 5th percentile value based on the percentile of the value.

Diagnoses were converted to ICD-10-CM, and the first three digits of the codes were extracted. We did not include diagnoses of mental and behavioural disorders, sexually transmitted infections, human immunodeficiency virus (HIV)-related disease, termination and abuse-related diagnoses, specific conditions originating in the perinatal period, congenital malformations, deformations, chromosomal abnormalities, and pre-defined rare disease according to the Rare Disease Management Act of Korea. The excluded ICD10-CM codes are uploaded as "icd10_excluded.csv" file.

Operation names were converted to the first four codes of ICD-10-PCS, representing section, body system, root operation, and body part, by manual mapping to reduce the risk of re-identification.

Ages were categorized into five-year intervals years, such that e.g. the age of 50 comprises the age ranges between 47.5 and 52.4. Furthermore, all timepoint variables (denoted as '*_time*') were de-identified by transforming them into relative time, referencing the initial admission time as the time zero.

Considering age (with 5-year interval) and sex as quasi-identifiers and ICD-10-PCS (the first 4 codes) as sensitive attributes, we calculated k-anonymity of 129, l-diversity of 58, and t-closeness of 0.049. Using the ARX Data Anonymization Tool ver. 3.9.1, open-source software for anonymising sensitive personal data, we conducted a re-identification risk analysis; the risk of all attacker models was lower than 0.002%. Given these results, we consider re-identification risk of the INSPIRE dataset to be very low.

Data Description

The INSPIRE dataset consists of six tables. Detailed definitions of the variables in the tables are listed in the 'schema.csv' file.

Each table can be connected using *subject_id*. A *subject_id* may be matched to one or more *hadm_ids*. A single *hadm_id* may be matched with one or more *op_ids*. While some changes were made to make it suitable for studying surgical patients, much of the structure was borrowed from the MIMIC dataset.

Operations

The ‘operations’ table consists of operation-related variables, including the demographic characteristics at the time of operation, operation or anesthesia time (*opstart_time*, *opend_time*, *anstart_time*, or *aneend_time*), type of operation presented as initial 5 characters of ICD-10-PCS (*icd10_pcs*), anesthesia type (*antype*), variables of cardiopulmonary bypass, postoperative ICU admission and discharge, or in-hospital mortality.

Only ICU admissions within 24 hours of surgery were included in the *icuin_time* and *icuout_time* columns to focus on ICU admissions immediately after surgery.

Diagnosis

The ‘diagnosis’ table includes all diagnoses claimed by a physician in the form of ICD-10-CM from the time zero to the discharge after the last operation, except for a set of pre-defined, sensitive diagnoses that needed to be removed. Only the first three digits of the ICD-10-CM code and the relative time of diagnosis were presented.

A full list of the ICD-10-CM codes and the description can be found on the Centers for Disease Control and Prevention (CDC) website [13].

Vitals

The ‘vitals’ table includes all intraoperative vital signs, anesthesia machine settings such as inspiratory flow of O₂ or concentration of anesthesia gas, or ventilatory parameters, like tidal volume or peak inspiratory pressure during operation. All variables were matched with *subject_id* and *op_id*, presented with value without the unit, and *chart_time* of 5-minute interval. Labels for the parameters are in the 'parameters.csv' file.

Ward_vitals

While the ‘vitals’ table included intraoperative vital signs, the ‘ward_vitals’ table included vital signs measured outside the operating room. From the time 0 to the time of discharge after the last operation, all measured vital signs were included. The *chart_time* was expressed in 5-minute intervals, with the imputation with the median values for variables measured shorter than 5 minutes. Labels for the parameters are in the 'parameters.csv' file.

Labs

Pre-defined laboratory variables were included in the ‘labs’ table with their value and *chart_time*. Laboratory results measured from 6 months before the time zero to 6 months after the last discharge were included. Labels for the parameters are in the 'parameters.csv' file.

Medications

The ‘medications’ table includes data on medications administrated between time 0 and the time of the last discharge. Information captured in the table includes *subject_id*, *chart_time* as the time of the drug administered, *drug_name* as the ingredient name, and *route* as the route of drug administered included in the ‘medications’ table. Fluid administrations such as balanced crystalloid, normal saline, or dextrose solution in general wards were not included. To avoid the risk of re-identification by using rarely administered medications, chemotherapy, immunotherapy, research drugs, and medications administered to less than 100 patients were excluded.

The World Health Organization(WHO) Anatomical Therapeutic Chemical (ATC) codes were manually mapped from the drug name and are provided in the *atc_code*, *atc_code2*, and *atc_code3*.

Usage Notes

INSPIRE data can be utilized to develop predictive models for post-operative complications. For example, we utilized INSPIRE data to develop a machine learning-based predictive model for 30-day postoperative mortality. The model had an AUROC of 0.944. The code has been uploaded to a public repository [14].

Anesthesia Practice

We use Hartmann or Plasma solution [15] for the main fluid. However, for chronic renal failure patients or in neurosurgery, we use normal saline. Sometimes, we use hydroxyethyl-starch for volume expansion. We don't use albumin except for treating hypo-albuminemia.

The primary vasopressors and doses are ephedrine 5 mg and phenylephrine 20-50 mcg. These drugs are frequently used intraoperatively. If these drugs do not treat the hypotension, we usually use norepinephrine. Other vasopressors and inotropics such as dopamine, dobutamine, epinephrine, vasopressin, milrinone, etc are often used in cardiac surgery.

We only use the target-controlled infusion (TCI) for propofol and remifentanil. Other drugs, such as inotropics and muscle relaxants, are infused with non-TCI.

We only use Sevoflurane and Desflurane for inhalation anesthesia. So, if you can find these parameters in the vitals table, the patients received inhalation anesthesia.

Release Notes

September 18, 2023

Version 0.1 (~52,000 cases)

October 4, 2023

Version 1.0 (~130,000 cases)

November 1, 2023

Version 1.1 (~130,000 cases)

We've categorized all the measurements into the 5th percentile values for greater anonymity. The details are described in the Anonymization subsection of the Methods section.

December 11, 2023

Version 1.2 (~130,000 cases)

We've updated the list of excluded diagnoses according to the Rare Disease Management Act of Korea. The excluded ICD10-CM codes are uploaded as "icd10_excluded.csv" file.

August 3, 2024

Version 1.3

We have corrected the icuin_time value for subject_id 189643811.

We have corrected the typo in the parameters table.

In the medication table, the compound medications separated by slash(/) within a single drug_name column were separated by drug_name, drug_name2, and drug_name3 columns.

We have added the World Health Organization(WHO) Anatomical Therapeutic Chemical (ATC) codes after manual mapping from the drug name and provided in the atc_code, atc_code2, and atc_code3 columns.

We have added the description of the criteria for the icuin_time and icuout_time as below.

Only ICU admissions within 24 hours of surgery were included in the icuin_time and icuout_time columns to focus on ICU admissions immediately after surgery.

We have added our routine anesthesia practice (fluid management, vasopressors, etc) in the usage note.

We have added an all-cause mortality column in the operations table.

Ethics

The acquisition and free disclosure of the data were approved by the Institutional Review Board of Seoul National University Hospital (H-2210-078-1368). Written informed consent was waived due to the anonymity of the data. Data collection was performed in accordance with relevant guidelines and regulations of the institutional Ethics Committee.

Acknowledgements

This research was supported by a grant of the Korea Health Technology R&D Project through the Korea Health Industry Development Institute (KHIDI), funded by the Ministry of Health & Welfare, Republic of Korea (grant number: HI21C1074)

Conflicts of Interest

None

References

1. Information of the Plasma solution A. https://www.health.kr/searchDrug/result_drug.asp?drug_cd=A11A1160A0270
2. Lee, H. C. et al. VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. Sci Data 9, 279, doi:10.1038/s41597-022-01411-5 (2022).
3. Vistisen, S. T., Pollard, T. J., Enevoldsen, J. & Scheeren, T. W. L. VitalDB: fostering collaboration in anaesthesia research. Br J Anaesth 127, 184-187, doi:10.1016/j.bja.2021.03.011 (2021).
4. Johnson, A. E. et al. MIMIC-III, a freely accessible critical care database. Sci Data 3, 160035, doi:10.1038/sdata.2016.35 (2016).
5. Bektas, M., Tuynman, J. B., Costa Pereira, J., Burchell, G. L. & van der Peet, D. L. Machine Learning Algorithms for Predicting Surgical Outcomes after Colorectal Surgery: A Systematic Review. World J Surg 46, 3100-3110, doi:10.1007/s00268-022-06728-1 (2022).
6. Tevis, S. E., Cobian, A. G., Truong, H. P., Craven, M. W. & Kennedy, G. D. Implications of Multiple Complications on the Postoperative Recovery of General Surgery Patients. Ann Surg 263, 1213-1218, doi:10.1097/SLA.0000000000001390 (2016).
7. Senanayake, S. et al. Machine learning in predicting graft failure following kidney transplantation: A systematic review of published predictive models. Int J Med Inform 130, 103957, doi:10.1016/j.ijmedinf.2019.103957 (2019).
8. Liao, A., Havidich, J. E., Onega, T. & Dutton, R. P. The National Anesthesia Clinical Outcomes Registry. Anesth Analg 121, 1604-1610, doi:10.1213/ANE.0000000000000895 (2015).
9. Moon, T. J. Light and shadows of the Korean healthcare system. J Korean Med Sci 27 Suppl, S3-6, doi:10.3346/jkms.2012.27.S.S3 (2012).
10. Penny-Dimri, J. C. et al. Machine learning to predict adverse outcomes after cardiac surgery: A systematic review and meta-analysis. J Card Surg 37, 3838-3845, doi:10.1111/jocs.16842 (2022).

11.

Fink, A. S. et al. The National Surgical Quality Improvement Program in non-veterans administration hospitals: initial demonstration of feasibility. Ann Surg 236, 344-353; discussion 353-344, doi:10.1097/00000658-200209000-00011 (2002).

12.

A full list of the ICD-10-CM codes. https://ftp.cdc.gov/pub/Health_Statistics/NCHS/Publications/ICD10CM/April-1-2023-Update/icd10cm-code%20descriptions-%20April%201%202023.zip

13.

Sample code of machine learning model for 30-day mortality after surgery. https://github.com/vitaldb/inspire/blob/main/gbm_mortality.py

14.

The Centers for Medicare and Medicaid Services and the National Center for Health Statistics, U. S. ICD-10-CM Official Guidelines for Coding and Reporting FY 2022, <https://www.cms.gov/files/document/fy-2022-icd-10-cm-coding-guidelines-updated-02012022.pdf> (2023).

15.

Steyerberg, E. W. & Harrell, F. E., Jr. Prediction models need appropriate internal, internal-external, and external validation. J Clin Epidemiol 69, 245-247, doi:10.1016/j.jclinepi.2015.04.005 (2016).

Contents

Parent Projects

INSPIRE, a publicly available research dataset for perioperative medicine was derived from:

- [VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients v1.0.0](#)

Please cite them when using this project.

Share



Access

Access Policy:
Only credentialed users who sign the DUA can access the files.

License (for files):
[Korea Credentialed Health Data License-1-0-0](#)

Data Use Agreement:
[Korea Credentialed Health Data Agreement-1-0-0](#)

Required training:
[CITI Data or Specimens Only Research](#)

Discovery

DOI (version 1.3):
<https://doi.org/10.13026/46m4-f655>

DOI (latest version):
<https://doi.org/10.13026/dv8j-5412>

Topics:

surgery

open dataset

multi-center

perioperative medicine

Project Website:
<https://inspire.or.kr>

Project Views

677

Current Version

1281

All Versions

Project Views by Unique Registered Users

View Details

Corresponding Author

You must be logged in to view the contact information.

Versions	
0.1	Sept. 19, 2023
1.0	Oct. 6, 2023
1.1	Nov. 3, 2023
1.2	Dec. 28, 2023
1.3	Aug. 12, 2024

Files

This is a restricted-access resource. To access the files, you must fulfill all of the following requirements:

- be a [credentialed user](#)
- complete required training:
 - [CITI Data or Specimens Only Research](#)You may submit your training [here](#).
- [sign the data use agreement](#) for the project



Maintained by the MIT Laboratory for Computational Physiology

Supported by the National Institute of Biomedical Imaging and Bioengineering (NIBIB), National Heart Lung and Blood Institute (NHLBI), and NIH Office of the Director under NIH grant numbers U24EB037545 and R01EB030362

Navigation

- [Discover Data](#)
- [Share Data](#)
- [About](#)
- [News](#)

Explore

[Data](#)

[Software](#)

[Challenges](#)

[Tutorials](#)



[Accessibility](#)